Homework 1(f) - Solutions

MT2005 - Probability and Statistics

February 24, 2025

1 Absolute Measures of Dispersion

1.1 Range

The range is calculated as:

$$\text{Range} = \max(x) - \min(x) \tag{1}$$

For House Size:

$$\text{Range} = 1800 - 1000 = 800 \tag{2}$$

For House Price:

$$\text{Range} = 400 - 220 = 180 \tag{3}$$

1.2 Interquartile Range (IQR)

For Number of Rooms, we calculate:

$$\text{IQR} = Q_3 - Q_1 \tag{4}$$

1.3 Mean Deviation

The mean deviation for House Size is:

$$\text{Mean Deviation} = \frac{\sum |x_i - \bar{x}|}{n} \tag{5}$$

1.4 Standard Deviation

The standard deviation is given by:

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}} \tag{6}$$

Calculations for House Size, Number of Rooms, and House Price follow.

2 Relative Measures of Dispersion

2.1 Coefficient of Range

$$\text{Coefficient of Range} = \frac{\max(x) - \min(x)}{\max(x) + \min(x)} \tag{7}$$

2.2 Coefficient of IQR

$$\text{Coefficient of IQR} = \frac{Q_3 - Q_1}{Q_3 + Q_1} \tag{8}$$

2.3 Coefficient of Mean Deviation

$$\text{Coefficient of Mean Deviation} = \frac{\text{Mean Deviation}}{\bar{x}} \tag{9}$$

2.4 Coefficient of Variation (CV%)

$$CV\% = \frac{\sigma}{\bar{x}} \times 100 \tag{10}$$

3 Correlation Coefficient

Using Pearson's formula:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} \tag{11}$$

4 Visualization

Python code for scatter plots:

```
import matplotlib.pyplot as plt import pandas as pd data = { 'House Size': [1200, 1500, 10
```

5 Feature Selection

Based on dispersion measures and correlation, House Size has a stronger impact on House Price.